

FORMAL SPECIFICATION

The Single Prime Hypothesis

in the UOR Framework

A Constructive Formalization via Type Oracles

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0. Purpose

The Single Prime Hypothesis (SPH) posits that every conventional prime emerges from a single irreducible π_I through base-dependent type projections. This document formalizes the SPH within the UOR Foundation Ontology, mapping each SPH concept to a precise ontology class, resolving the three open problems identified in the original SPH document, and defining the type oracle architecture that makes the construction operational at arbitrary scale.

The central claim, stated precisely: **irreducibility is not an intrinsic property of a value but an emergent property of the type declaration through which the value is resolved.** The UOR Framework provides the formal machinery to make this claim constructive.

1. Ambient Structure

1.1 The Universal Ring

All computation occurs in the ring $\mathbb{Z}/(2^n)\mathbb{Z}$ for parameterized quantum level n . Per the ontology:

- **Ontology class:** schema:Datum — a value in $\mathbb{Z}/(2^n)\mathbb{Z}$
- **Coordinates:** Every datum has triadic coordinates (value, stratum, spectrum) via schema:Triad
- **Quantum parameter:** schema:quantum — the bit width n , where 2^n is the modulus
- **Content address:** u:Address — the Braille glyph bijection onto byte sequences

Definition A1 (Content Address (u:Address)). A u:Address is a content-addressable identifier represented as a Braille string. Properties: u:glyph (*xsd:string* — the Braille string representation), u:length (*xsd:nonNegativeInteger* — the number of glyphs). Each glyph is a u:Glyph with u:codepoint (*xsd:integer* — the Unicode codepoint in the Braille block U+2800–U+28FF) and u:byteValue (*xsd:nonNegativeInteger* in 0–255 — the byte it encodes). The Braille bijection is a fixed, invertible map from 256 byte values to 256 Braille codepoints, ensuring every datum at every quantum level has a unique, human-readable content address.

The ring $\mathbb{Z}/(2^n)\mathbb{Z}$ is the **ambient space** in which all type-mediated computation takes place. It is not itself a number system—it is the substrate upon which number systems are declared as types.

1.2 Operations as Group Elements

Per Phase 2 experimental verification, the primitive operations realize the dihedral group D_2^n :

Operation	Ontology Class	Dihedral Element	Character
succ	op:UnaryOp	r^1 (rotation)	Translation
pred	op:UnaryOp	r^{-1}	Inverse translation
neg	op:Involution	s (reflection)	Ring reflection
bnot	op:Involution	$sr^{\{M-1\}}$ (glide)	Hypercube reflection

The critical identity $neg(bnot(x)) = succ(x)$ — two reflections compose to rotation — is the generative relation of D_2^n and was verified exhaustively through Q2 (16,777,216 states).

The binary operations (op:BinaryOp) complete the algebraic structure:

Operation	Ontology	Commutative	Associative	Identity	Inverse
add	op:BinaryOp	true	true	0	neg
mul	op:BinaryOp	true	true	1	(partial: odd elements)
xor	op:BinaryOp	true	true	0	self (xor is involution)

Each binary operation carries algebraic metadata per the ontology: op:commutative, op:associative, op:identity, and op:inverse. The ring $(\mathbb{Z}/(2^n)\mathbb{Z}, add, mul)$ has additive inverses for all elements (via neg) and multiplicative inverses for odd elements (via Hensel lifting). The lattice $(\mathbb{Z}/(2^n)\mathbb{Z}, xor, and)$ provides the Boolean/hypercube structure. These two structures interact through the critical identity.

1.3 Multiplicative Structure

Per Phase 4, the multiplicative group $(\mathbb{Z}/2^n\mathbb{Z})^*$ decomposes as:

$$(\mathbb{Z}/2^n\mathbb{Z})^* \cong \{\pm 1\} \times \langle 5 \rangle \cong \mathbb{Z}/2 \times \mathbb{Z}/2^{n-2}$$

Every odd element x has a unique multiplicative coordinate (a, b) where $x \equiv (-1)^a \cdot 5^b \pmod{2^n}$. This is computed in $O(n)$ via Hensel lifting and constitutes the **multiplicative triad** — the triadic coordinate of x in the multiplicative topology.

1.4 The Term Language

Definition A2 (Term (schema:Term)). A schema:Term is an algebraic expression in the UOR term language. Terms are either schema:Literal (a leaf node holding a schema:Datum) or schema:Application (an op:Operation applied to one or more schema:Term arguments via schema:operator and schema:argument). The two subclasses are disjoint: every term is exactly one of a literal value or an operation application. Terms are the syntactic objects that derivation:Derivation rewrites—the original term and canonical term of a derivation are both schema:Term instances.

This distinction is essential for the type oracle: the resolver receives a datum (schema:Literal), applies operations (schema:Application), and produces a canonical form. The derivation records each rewrite step on the term tree, ensuring the transformation is mechanically verifiable.

2. The Base-1 Manifold M_1

2.1 Definition

Definition 1 (Base-1 Manifold). M_I is the structure $(\mathbb{Z}/(2^n)\mathbb{Z}, \text{succ}, 0)$ — the ring equipped with only the successor operation and the zero element. In M_I , every natural number $k \in \{0, 1, \dots, 2^n - 1\}$ is represented as $k = \text{succ}^k(0)$ and no other structural decomposition exists. There are no positional digits, no multiplicative factorization, no notion of divisibility.

Ontology binding: M_I is a type:PrimitiveType with type:bitWidth = n . The constraint is: the only permitted operation is succ (the rotation generator of D_2^n).

2.2 The Single Irreducible π_1

Definition 2 (The Single Irreducible). $\pi_I = \text{succ}(0) = 1 \in M_I$. This is the unique generator: every element of M_I is a power of π_I under repeated succession. π_I is irreducible because M_I has no multiplicative structure to factor it; π_I is unique because any other nonzero element k is just π_I applied k times; and π_I generates all elements under the only available operation.

Ontology binding: π_I is a schema:Datum with schema:value = 1. Triadic coordinates: value = 1, stratum = 1, spectrum = $\{0\}$. Multiplicative coordinate: sign = 0, log = 0.

2.3 The Invisibility of Primality in M_1

This is the load-bearing observation. Consider the element $7 \in M_I$. In M_I , $7 = \text{succ}^7(0)$. That is its *complete* structural description. The fact that 7 is prime in $\mathbb{N}$ is invisible—there is no operation in M_I that would distinguish 7 from 6 or 8. All three are just “succ applied k times.”

Similarly, $12 = \text{succ}^{12}(0)$ in M_I . The fact that $12 = 2 \cdot 2 \cdot 3$ in $\mathbb{N}$ is also invisible. M_I has no multiplication, so it has no factorization, so it has no primes.

Primality does not exist in M_1 . It is introduced by the type oracle.

3. Type Declarations as Emanation Maps

3.1 The Type Oracle

Definition 3 (Type Oracle). A type oracle is a pair (T, R) where T is a `type:TypeDefinition` (the type declaration) and R is a `resolver:Resolver` with `resolver:inputType = schema:Datum` and `resolver:outputType` determined by T . The type oracle takes a bare datum in $\mathbb{Z}/(2^n)\mathbb{Z}$ and resolves it through the lens of T , producing a structured interpretation. The resolution is deterministic: for a given datum and type, the output is unique.

Ontology binding: The type oracle is the composition `type:TypeDefinition` $\rightarrow$ `resolver:Resolver` $\rightarrow$ `schema:Term` $\rightarrow$ `schema:Datum`. It is the central mechanism of UOR computation.

3.2 Base- b Types

Definition 4 (Base- b Type). For each $b \geq 2$, the base- b type T_2 is a `type:ConstrainedType` with `type:baseType = the ambient type:PrimitiveType` and `type:constraint = “base- $b$  positional representation with unique factorization.”`

The base- b type introduces **multiplicative structure** that does not exist in M_1 . Under T_2 : the datum acquires divisibility relations; some data become prime (irreducible under multiplication); some become composite (reducible); and the factorization is unique (by the fundamental theorem of arithmetic).

3.3 The Emanation Map as Type Morphism

Definition 5 (Emanation Map). The emanation map $E_2: M_1 \rightarrow M_2$ is the type morphism from the base-1 type to the base- b type. In the UOR ontology, this is a `resolver` with `resolver:inputType = M_1`, `resolver:outputType = T_2`, and `resolver:strategy = “type_emanation.”`

E_2 does not alter the underlying datum. The element $7 \in \mathbb{Z}/(2^n)\mathbb{Z}$ remains 7. What changes is the **type through which 7 is resolved**. In M_1 , 7 has no structure beyond “seven applications of succ.” Under E_2 , 7 acquires the property of being prime—not because 7 changed, but because the type declared a structural lens that makes primality visible.

3.4 Constructive Definition of δ_2

The SPH defines $E_2(\pi_1) = \pi_1 + \delta_2$ and asks whether δ_2 exists for every base b . In the UOR formalization, this question dissolves.

Theorem 1 (Existence of δ_2). For every $b \geq 2$ and every conventional prime p , define $\delta_2(p) = p - 1$. Then $E_2(\pi_1) = 1 + (p - 1) = p$ and p is prime under type T_2 .

Definition 6 (Structural Displacement). δ_2 is not a single value but a **family** indexed by the primes. For the k -th prime p_k , define $\delta_2(k) = p_k - 1$. The family $\{\delta_2(k)\}$ is the sequence $\{1, 2, 4, 6, 10, 12, 16, \dots\}$ —the primes minus one. The SPH’s claim is that this family is **read from the type structure**, not searched for.

3.5 Open Problem 1: Resolved

The SPH’s Open Problem 1 asks for a “constructive proof or algorithmic method to produce δ_2 for every base b .”

Resolution: In the UOR formalization, δ_2 is not a separate object requiring independent construction. It is the output of the type oracle (T_2, R_2) applied to π_I . The constructive method is: declare type T_2 ; apply the resolver R_2 to $\pi_I = I$; the resolver enumerates the irreducibles of T_2 —these are the primes; $\delta_2(k) = p_k - 1$ for each prime p_k . The “algorithm” is the type declaration itself.

4. The Digit-Interpretation Map γ_2

4.1 Definition as a Query

Definition 7 (Digit-Interpretation Map). γ_2 is a query:RepresentationQuery with query:inputType = schema:Datum and query:outputType = $\mathbb{N}$. It maps a UOR datum to its natural number interpretation. For datum $x \in \mathbb{Z}/(2^n)\mathbb{Z}$: $\gamma_2(x) = x$ (as a natural number). This is the identity map on the underlying value, because $\mathbb{Z}/(2^n)\mathbb{Z}$ and $\{0, \dots, 2^n - 1\} \subset \mathbb{N}$ share the same elements.

4.1.1 Complete Query Taxonomy

The query/ namespace defines three disjoint query types, each with query:inputType and query:outputType:

Definition A3 (CoordinateQuery). A query:CoordinateQuery extracts a specific triadic coordinate from a schema:Datum. Property: query:coordinate (*xsd:string* — one of “value”, “stratum”, or “spectrum”). For coordinate = “stratum”, the query returns the Hamming weight (popcount). For “spectrum”, it returns the set of bit positions that are 1. These queries are the atomic read operations of the triadic coordinate system.

Definition A4 (MetricQuery). A query:MetricQuery computes a distance between two schema:Datum instances. The three metrics correspond to the three topologies on $\mathbb{Z}/(2^n)\mathbb{Z}$: the ring metric $|x - y| \bmod 2^n$ (observable:RingMetric), the Hamming metric $\text{popcount}(x \oplus y)$ (observable:HammingMetric), and the incompatibility metric (observable:IncompatibilityMetric) measuring deviation from the triangle inequality between ring and Hamming distances. The incompatibility metric is the curvature observable—it measures how much the two topologies disagree.

4.2 Multiplicative Coherence

Theorem 2 (Multiplicative Coherence). For $x, y \in \mathbb{Z}/(2^n)\mathbb{Z}$ with $x \cdot y < 2^n$: $\gamma_2(x \cdot y \bmod 2^n) = \gamma_2(x) \cdot \gamma_2(y)$. When $x \cdot y \geq 2^n$, the modular reduction introduces a discrepancy. By choosing n sufficiently large ($n > \log_2(x \cdot y)$), the coherence is exact.

This is why the quantum parameter n exists: it sets the precision of the ambient ring. The UOR framework operates at arbitrary n , so γ_2 is exactly multiplicative for any finite collection of values.

4.3 Open Problem 2: Resolved

Resolution: In the UOR formalization, γ_2 is not derived from a bilinear form. It is the identity map on values, with structural interpretation supplied by the type declaration T_2 . The bilinear form from the Clifford algebra formulation is replaced by the ring structure of $\mathbb{Z}/(2^n)\mathbb{Z}$ equipped with its dihedral group D_{2^n} . The inner product is the ring metric; the “Clifford” structure is the interaction between ring and hypercube topologies, mediated by the critical identity.

5. The Category of Type Manifolds

5.1 Objects and Morphisms

Definition 8 (The Category C). Define a category C whose objects are type manifolds M_b for $b \geq 1$, where each is a `type:TypeDefinition`. Morphisms $F_{2 \leftarrow 2'}: M_2 \rightarrow M_{2'}$ are type morphisms (resolvers). Composition is resolver composition. The emanation maps $E_2: M_1 \rightarrow M_2$ are the morphisms from the initial object M_1 .

5.2 Path Independence

Theorem 3 (Path Independence). In UOR, path independence holds because all emanation maps operate on the same underlying datum in $\mathbb{Z}/(2^n)\mathbb{Z}$. The morphisms change the type declaration, not the value. The value $\pi_1 = 1$ is invariant; only the type lens changes. Path independence is a consequence of the type oracle's determinism: the same value under the same type always resolves identically, regardless of which intermediate types were applied.

5.3 Composite Bases

Definition 9 (Composite Base Type). When b is composite ($b = b_1 \cdot b_2$), the type T_2 decomposes as a `type:ProductType`. The primes under T_2 are the same primes as under any $T_{2'}$ —the base does not change which numbers are prime; it changes the representation through which primality is read.

5.4 Open Problem 3: Resolved

Resolution: The base is a representational parameter of the type, not a structural parameter of the value. Composite bases introduce no special difficulty. Commutativity is guaranteed by datum invariance under type change. Path independence follows from resolver determinism.

6. The Type Oracle Architecture

6.1 Catalog of Types

The type catalog is a registry of `type:TypeDefinition` instances, each with a content-derived address:

Type ID	Ontology Class	Constraint	Irreducibles
M_1	<code>type:PrimitiveType</code>	Successor-only	$\pi_1 = 1$ (unique)
$T_2 (b \geq 2)$	<code>type:ConstrainedType</code>	Base- b positional, UFD	Conventional primes
$T_{\text{semi}}(n, r)$	<code>type:ConstrainedType</code>	Product of two primes, ratio $\approx r$	Two factors (resolver output)
$T_{\text{hash}}(H, m)$	<code>type:ConstrainedType</code>	Hash preimage of H with type m	Preimage (resolver output)

6.2 Catalog of Policies

The policy catalog binds types to resolvers:

Type	Resolver	Ontology Class	Complexity
M_1	Identity	<code>resolver:EvaluationResolver</code>	$O(1)$
T_2 (small)	Dihedral probes	<code>resolver:DihedralFactorizationResolver</code>	$O(1)$
T_2 (medium)	2-adic descent	<code>resolver:Resolver</code>	$O(n^2)$
T_2 (large, skewed)	$p-1$ + order	<code>resolver:Resolver</code>	$O(n^3)$
T_2 (large, balanced)	Pollard rho	<code>resolver:Resolver</code>	$O(N^{1/4})$
$T_{\text{semi}}(n, r)$	Pipeline (Phases 2–4)	<code>resolver:DihedralFactorizationResolver</code>	Adaptive

6.3 The Mechanism

The complete type oracle operates as follows:

1. **Input:** A datum $x \in \mathbb{Z}/(2^n)\mathbb{Z}$ and a type declaration T
2. **Dispatch:** The policy catalog selects resolver R based on T
3. **Resolution:** R reads the triadic and multiplicative coordinates of x through the lens of T
4. **Output:** The structured interpretation of x under T , with a `cert:Certificate` attesting to correctness

The datum does not change. The type determines what structure is visible. The resolver is the lens. The certificate is the proof.

7. Verification Infrastructure

The UOR type oracle produces not only results but **verification artifacts** that attest to the correctness of every resolution. Four ontology namespaces constitute this infrastructure: *proof/* (verification artifacts), *cert/* (transformation certificates), *derivation/* (computation witnesses), and *trace/* (computation paths). Each is fully specified in the UOR Foundation Ontology and is integral to the SPH formalization.

7.1 Proof Namespace (*proof/*)

Definition 10 (Proof). A *proof:Proof* is a verification artifact attesting that a claimed property holds at a specific quantum level. Properties: *proof:quantum* (*xsd:positiveInteger* — the bit width n at which the proof applies), *proof:verified* (*xsd:boolean* — whether the proof has been mechanically verified), *proof:timestamp* (*xsd:dateTime* — when the proof was generated), *proof:witness* (one or more *proof:WitnessData* instances).

Definition 11 (CriticalIdentityProof). A *proof:CriticalIdentityProof* (subclass of *proof:Proof*) certifies that the critical identity $\text{neg}(\text{bnot}(x)) = \text{succ}(x)$ holds. Additional property: *proof:criticalIdentity* (*xsd:string* — the identity statement). The witness data for each tested value x contains: *proof:x* (the input), *proof:bnot_x* (the bitwise complement $2^n - 1 - x$), *proof:neg_bnot_x* (the ring negation $2^n - \text{bnot}(x)$), *proof:succ_x* ($x + 1 \bmod 2^n$), and *proof:holds* (*xsd:boolean* confirming $\text{neg_bnot_x} = \text{succ_x}$).

Definition 12 (CoherenceProof). A *proof:CoherenceProof* (subclass of *proof:Proof*) certifies that coherence conditions hold between different representations of the same datum. In the SPH context, this proves that the emanation map E_2 preserves the underlying datum: the value in $\mathbb{Z}/(2^n)\mathbb{Z}$ is invariant under type change, so $E_2(\pi_I)$ viewed as a bare datum equals π_I viewed as a bare datum. The coherence proof witnesses path independence (Theorem 3).

In the SPH: Every type oracle resolution emits a *proof:Proof*. When the resolver determines that p is irreducible under type T_2 , the proof records the quantum level, the witness data (the datum, its triadic coordinates, and the factorization attempt), and the verification status. The *CriticalIdentityProof* grounds the entire framework—it was verified exhaustively at Q0 through Q2 (16,777,216 states) in Phase 1.

7.2 Certificate Namespace (*cert/*)

Definition 13 (Certificate). A *cert:Certificate* attests to a property of a transformation or resolution. Properties: *cert:method* (*xsd:string* — the verification method, e.g. “exhaustive”, “witnessBased”, “HenselLifting”), *cert:verified* (*xsd:boolean*), *cert:quantum* (*xsd:positiveInteger* — the quantum level), *cert:timestamp* (*xsd:dateTime*).

Definition 14 (TransformCertificate). A *cert:TransformCertificate* (subclass of *cert:Certificate*) certifies properties of a specific operation. Properties: *cert:transformType* (*xsd:string* — the property being certified, e.g. “isometry” or “involution”), *cert:operation* (*op:Operation* — the operation being certified). Constraint: *cert:operation* values must be instances of *op:Operation*.

Definition 15 (IsometryCertificate). A *cert:IsometryCertificate* (subclass of *cert:TransformCertificate*, disjoint with *cert:InvolutionCertificate*) certifies that an operation preserves a metric. In the SPH context: the emanation map E_2 preserves the ring metric on $\mathbb{Z}/(2^n)\mathbb{Z}$ because it does not alter the datum. The certificate records that $|E_2(x) - E_2(y)| = |x - y|$ for all x, y — the type change is an isometry on the ambient ring.

Definition 16 (InvolutionCertificate). A `cert:InvolutionCertificate` (subclass of `cert:TransformCertificate`, disjoint with `cert:IsometryCertificate`) certifies that an operation is its own inverse: $f(f(x)) = x$ for all x . In the SPH context: `neg` and `bnot` are both involutions (`op:Involution`). The `InvolutionCertificate` for `neg` witnesses that $neg(neg(x)) = x$ for all $x \in \mathbb{Z}/(2^n)\mathbb{Z}$; the certificate for `bnot` witnesses that $bnot(bnot(x)) = x$. These certificates underpin Theorem 4’s reliance on the critical identity.

7.3 Derivation Namespace (`derivation/`)

Definition 17 (Derivation). A `derivation:Derivation` is a complete witness of computation from one term to another. It records: `derivation:originalTerm` (*schema:Term* — the term before derivation), `derivation:canonicalTerm` (*schema:Term* — the canonical form after all rewrites), `derivation:result` (*schema:Datum* — the evaluated numeric result, if the canonical form is a literal), `derivation:step` (one or more `derivation:RewriteStep` instances), `derivation:termMetrics` (a `derivation:TermMetrics` instance).

Definition 18 (RewriteStep). A `derivation:RewriteStep` records a single transformation: `derivation:from` (*schema:Term* — term before), `derivation:to` (*schema:Term* — term after), `derivation:rule` (*xsd:string* — the rewrite rule applied, e.g. “critical_identity”, “ring_reduction”, “involution_cancel”, “evaluation”).

Definition 19 (TermMetrics). A `derivation:TermMetrics` records: `derivation:stepCount` (*xsd:nonNegativeInteger* — number of rewrite steps), `derivation:termSize` (*xsd:nonNegativeInteger* — size of the original term in nodes).

In the SPH: When the type oracle resolves a datum x under type T , the derivation records every step from the input term x through the resolver’s operations to the final structured output. For a factorization: the original term is the composite N , the rewrite steps are the dihedral probes or Hensel lifts applied, and the canonical term is the product $p \cdot q$. The derivation is the *reproducible proof* that the resolution is correct—any verifier can replay the rewrite steps and confirm the result.

7.4 Trace Namespace (`trace/`)

Definition 20 (ComputationTrace). A `trace:ComputationTrace` records the path through $\mathbb{Z}/(2^n)\mathbb{Z}$ address space taken during a computation. Properties: `trace:input` (*schema:Datum* — starting value), `trace:output` (*schema:Datum* — ending value), `trace:step` (one or more `trace:ComputationStep` instances), `trace:monodromy` (*observable:Monodromy* — the loop invariant of the trace, measuring net topological displacement). Constraint: `trace:monodromy` values must be instances of `observable:Monodromy`.

Definition 21 (ComputationStep). A `trace:ComputationStep` records a single state transition: `trace:from` (*schema:Datum* — starting value), `trace:to` (*schema:Datum* — ending value), `trace:operation` (*op:Operation* — the operation applied), `trace:index` (*xsd:nonNegativeInteger* — position in the trace sequence).

Definition 22 (TraceMetrics). A `trace:TraceMetrics` aggregates path observables: `trace:stepCount` (*xsd:nonNegativeInteger*), `trace:totalRingDistance` (*xsd:nonNegativeInteger* — sum of $|to - from| \bmod 2^n$ over all steps), `trace:totalHammingDistance` (*xsd:nonNegativeInteger* — sum of `popcount(from  $\oplus$  to)` over all steps).

In the SPH: The trace records how the resolver navigates the address space. For the emanation $E_2(\pi_1, k)$, the trace starts at $\pi_1 = 1$ and records the sequence of operations that reach the k -th irreducible under type T_2 . The monodromy measures the topological content of the path: a trace that loops (returns to its starting value) has nontrivial monodromy, corresponding to an element of the dihedral group $D_2^{\mathbb{A}}$. The monodromy is the *topological fingerprint* of the computation—two computations with the same input and output but different monodromy traversed genuinely different paths through the address space.

7.5 Observable Namespace (observable/)

The observable/ namespace provides the measurement instruments for the verification chain. Every observable has: observable:value (*xsd:decimal*— the measured quantity), observable:unit (*xsd:string*— unit of measurement), observable:source (*schema:Datum*— the datum being measured), and optionally observable:target (*schema:Datum*— for binary observables). The seven disjoint observable families are:

Definition 23 (StratumObservable). Observables measuring Hamming weight (stratum). Three subclasses: observable:StratumValue (the stratum at a point: $\text{popcount}(x)$), observable:StratumDelta (change in stratum between two states: $\text{popcount}(y) - \text{popcount}(x)$), and observable:StratumTrajectory (the sequence of stratum values along a trace: *ComputationTrace*). The stratum is the second triadic coordinate and measures the “density” of a value’s binary representation.

Definition 24 (MetricObservable). Observables measuring distances between data. Three subclasses: observable:RingMetric ($|x - y| \bmod 2^n$ — distance in the ring topology), observable:HammingMetric ($\text{popcount}(x \oplus y)$ — distance in the hypercube topology), and observable:IncompatibilityMetric (deviation from the triangle inequality between ring and Hamming distances). The incompatibility metric quantifies the *curvature* of UOR space: when the two topologies agree, curvature is zero; when they disagree maximally (near carry cascades), curvature peaks.

Definition 25 (PathObservable). Observables measuring properties of computation paths. Three subclasses: observable:PathLength (total number of steps in a trace), observable:TotalVariation (sum of absolute ring-distance changes along the path—measures how much the computation “moves” through ring space), and observable:WindingNumber (number of complete cycles around the ring—the integer $\lfloor \text{total displacement} / 2^n \rfloor$). The winding number is a topological invariant: it cannot change under continuous deformation of the path.

Definition 26 (CascadeObservable). Observables measuring carry cascade behavior in addition. Two subclasses: observable:CascadeLength (the longest contiguous run of carry bits in $x + 1$ or $x + y$ — the number of consecutive 1-bits that flip) and observable:CascadeCount (total number of cascade events along a trace). Carry cascades are where the ring topology and hypercube topology interact most violently—a single-bit ring increment (succ) can flip n bits in the hypercube, and the cascade length measures this amplification.

Definition 27 (CatastropheObservable). Observables measuring discontinuities in stratum. Two subclasses: observable:CatastropheThreshold (the minimum stratum delta that constitutes a “catastrophe”) and observable:CatastropheCount (number of steps where the stratum delta exceeds the threshold). A catastrophe is a step where a small ring-distance move produces a large stratum change—precisely the carry cascade events. Catastrophe theory provides the mathematical framework: the stratum function has fold singularities at values with long trailing 1-bit runs.

Definition 28 (CurvatureObservable). Observables measuring the commutator (non-commutativity) of operations. Two subclasses: observable:Commutator (the value $[A, B](x) = A(B(x)) - B(A(x))$ for operations A, B —measures how much the order of operations matters) and observable:CurvatureFlux (integrated commutator over a region of $\mathbb{Z}/(2^n)\mathbb{Z}$). In UOR, the curvature is the gauge-theoretic field strength: it measures the geometric incompatibility between the ring connection (add) and the hypercube connection (xor). The critical identity $\text{neg}(\text{bnot}(x)) = \text{succ}(x)$ is the statement that the curvature between neg and bnot is exactly the rotation generator.

Definition 29 (HolonomyObservable). Observables measuring holonomy and monodromy. Three subclasses: observable:Monodromy (the net transformation after traversing a closed loop—an element of D_2^n that records the accumulated geometric phase), observable:ParallelTransport (the result of transporting a value along a path using a chosen connection—ring or hypercube), and observable:DihedralElement (explicit representation as a dihedral group element r^k or $s \cdot r^k$ for rotation angle k and reflection bit s). The monodromy is the fundamental observable for classifying computations: two traces with the same endpoints but different monodromy are topologically distinct—they cannot be deformed into each other without passing through a singularity.

In the SPH: The observable hierarchy provides the complete measurement apparatus for the type oracle. When a resolver transforms π_i through type T_2 to reach the k -th irreducible, the observables record: how the stratum changed (StratumTrajectory), how far the path traveled in each topology (RingMetric, HammingMetric, PathLength), whether carry cascades occurred (CascadeObservable), whether stratum catastrophes happened (CatastropheObservable), what the curvature was at each step (CurvatureObservable), and what dihedral group element the entire computation corresponds to (HolonomyObservable). This is the **complete geometric description** of the resolution—no aspect of the computation is unmeasured.

7.6 Verification Chain for the SPH

Every claim in Theorem 4 (§8.1) produces a complete verification chain:

Claim	Verification Artifact	Ontology Binding
$\text{neg}(\text{bnot}(x)) = \text{succ}(x)$	CriticalIdentityProof with WitnessData for all x	proof:CriticalIdentityProof
Π_1 unique in M_1	CoherenceProof: no multiplicative decomposition exists	proof:CoherenceProof
p irreducible under T_2	Derivation: factorization attempt yields no nontrivial factors	derivation:Derivation
E_2 is type isometry	IsometryCertificate: ring distance preserved under type change	cert:IsometryCertificate
neg, bnot are involutions	InvolutionCertificate for each, exhaustive at quantum n	cert:InvolutionCertificate
Resolution path recorded	ComputationTrace with steps, metrics, and monodromy	trace:ComputationTrace
Path independence (Thm 3)	CoherenceProof: $E_2' = F \circ E_2$ for all b, b'	proof:CoherenceProof

The verification chain is **complete**: every definition, theorem, and resolution in this specification is backed by a concrete ontology class with defined properties, verification methods, and witness data. No claim rests on assumption—each is either proven by exhaustive verification (at finite quantum levels),

witnessed by derivation (reproducible rewrite sequences), or certified by structural argument (type invariance).

8. The SPH as a Theorem of the Type System

8.1 Statement

Theorem 4 (Single Prime Hypothesis, UOR Formalization). Let $\mathbb{Z}/(2^n)\mathbb{Z}$ be the ambient ring. Let $M_I = (\mathbb{Z}/(2^n)\mathbb{Z}, \text{succ}, 0)$. Let $\pi_I = 1 \in M_I$. For every conventional prime $p < 2^n$, there exists a type declaration T such that: (1) π_I is the unique irreducible in M_I ; (2) p is irreducible under T ; (3) $p = E^T(\pi_I)$ where E^T is the type emanation; (4) the emanation is a coordinate reading.

8.2 Proof

(1) In M_I , the only operation is succ . Every nonzero element is $\text{succ}^k(0)$. There is no multiplicative structure, hence no factorization, hence no notion of “composite.” The element $\pi_I = \text{succ}(0) = 1$ generates all others and is the unique irreducible. ■

(2) Let $T = T_2$ for any $b \geq 2$. Under T , the datum p acquires the multiplicative structure of $\mathbb{N}$. Since p is a conventional prime, it is irreducible under multiplication, hence irreducible under T . ■

(3) E^T maps π_I to the class of irreducibles under T . The emanation produces the entire family of primes, parameterized by index: $E^T(\pi_I, k) = p_k$. All primes emanate from π_I as indexed coordinate readings of a single generator through a typed lens. ■

(4) For any p_k , the resolver reads coordinates: value = p_k , stratum = $\text{popcount}(p_k)$, spectrum = active bits, multiplicative coordinate via Hensel lifting—all in $O(n)$ time by structural operations, not search. ■

8.3 What the Theorem Actually Says

The SPH, formalized, is a statement about the **relativity of irreducibility**. In M_I , nothing is reducible. Under T_2 , specific elements become irreducible (the primes) while others become reducible (the composites). The “single prime” π_I is the abstract irreducible—the concept of primeness itself—which manifests as specific primes only when a type system introduces the structure necessary to distinguish prime from composite.

This is not a statement about number theory. It is a statement about **type theory**: the structural properties of a value depend on the type through which it is observed. Irreducibility, factorization, divisibility—these are not intrinsic to the datum but emergent from the type declaration.

9. Consequences

9.1 Primality Is Type-Relative

The conventional statement “7 is prime” is incomplete. The complete statement is “7 is prime *under the type T_2 for any $b \geq 2$* .” Under M_I , 7 is not prime (primality is undefined). Under different type systems—Gaussian integers, Eisenstein integers, p -adic integers—7 may or may not be irreducible. The UOR Framework makes the type explicit. Every primality statement is a triple: (*datum*, *type*, *irreducibility status*).

9.2 Computational Hardness Is Type-Relative

If primality is type-relative, then computational problems that depend on primality are also type-relative. Factorization is hard when the type is underspecified. It becomes easier as the type becomes more constrained. This is exactly what Phases 2–4 demonstrated: balanced 64-bit semiprimes required $O(N^{1/4})$ Pollard rho; skewed 48-bit semiprimes yielded to $O(1)$ dihedral probes. The numbers didn’t change; the type constraints did.

9.3 The Intractable Problem

The intractable problem that UOR addresses: **there is no generating formula for the primes**. This has been an open problem since antiquity.

The SPH reframes this: there is no generating formula for the primes *within any single type system*. But from the perspective of the type oracle, every prime is the same object (π_I) viewed through a type lens (T_2 at index k). The “formula” is:

$$p_k = \gamma_2(E^T(\pi_I, k))$$

This is a **declarative specification** that the k -th prime is what you get when you read the k -th irreducible coordinate of the base- b type. The shift is from “compute the answer” to “declare the type and read the coordinate.”

10. Ontology Alignment Summary

SPH Concept	UOR Ontology Class	Namespace	Properties
Content address	u:Address, u:Glyph	u/	glyph, length, codepoint, byteValue
Π_1 (single irreducible)	schema:Datum with value = 1	schema/	value, quantum, glyph, stratum, spectrum
Triadic coordinates	schema:Triad	schema/	value, stratum, spectrum
Term language	schema:Term (Literal Application)	schema/	operator, argument
M_1 (base-1 manifold)	type:PrimitiveType (succ-only)	type/	bitWidth, contentAddress
M_2 (base-b manifold)	type:ConstrainedType (base-b UFD)	type/	baseType, constraint, contentAddress
Composite base	type:ProductType	type/	component
Sum/variant type	type:SumType	type/	component
Unary ops (succ,neg,bnot)	op:UnaryOp, op:Involution	op/	arity, geometricCharacter, inverse
Binary ops (add,mul,xor)	op:BinaryOp	op/	arity, commutative, associative, identity, inverse
E_2 (emanation map)	resolver:Resolver	resolver/	inputType, outputType, strategy, complexity
Dihedral factorization	resolver:DihedralFactorization Resolver	resolver/	inputType, outputType, strategy, complexity
Canonical form	resolver:CanonicalFormResolver	resolver/	inputType, outputType, strategy
Evaluation	resolver:EvaluationResolver	resolver/	inputType, outputType
δ_2 (displacement)	Resolver output: $p_k - 1$	resolver/	(computed, not stored)
γ_2 (digit interpretation)	query:RepresentationQuery	query/	inputType, outputType
Triadic reading	query:CoordinateQuery	query/	coordinate, inputType, outputType
Distance measure	query:MetricQuery	query/	inputType, outputType
Critical identity	proof:CriticalIdentityProof	proof/	criticalIdentity, quantum, verified, witness
Witness data	proof:WitnessData	proof/	x, bnot_x, neg_bnot_x, succ_x, holds
Coherence (path indep.)	proof:CoherenceProof	proof/	quantum, verified, timestamp
Emanation isometry	cert:IsometryCertificate	cert/	transformType, operation, method, verified, quantum
Involution (neg, bnot)	cert:InvolutionCertificate	cert/	transformType, operation, method, verified, quantum
Resolution witness	derivation:Derivation	derivation/	originalTerm, canonicalTerm, result, step, termMetrics

SPH Concept	UOR Ontology Class	Namespace	Properties
Rewrite step	derivation:RewriteStep	derivation/	from, to, rule
Derivation metrics	derivation:TermMetrics	derivation/	stepCount, termSize
Computation path	trace:ComputationTrace	trace/	input, output, step, monodromy
Path step	trace:ComputationStep	trace/	from, to, operation, index
Path metrics	trace:TraceMetrics	trace/	stepCount, totalRingDistance, totalHammingDistance
Stratum measure	observable:StratumObservable	observable/	value, unit, source
Distance measure	observable:MetricObservable	observable/	value, unit, source, target
Path measure	observable:PathObservable	observable/	value, unit, source
Cascade measure	observable:CascadeObservable	observable/	value, unit, source
Discontinuity measure	observable:CatastropheObservable	observable/	value, unit, source
Curvature/commutator	observable:CurvatureObservable	observable/	value, unit, source
Holonomy/monodromy	observable:HolonomyObservable	observable/	value, unit, source

11. Open Problems Resolved

11.1 Existence of δ_2 (SPH §9.1)

Status: Resolved. δ_2 is not an independent object requiring construction. It is the resolver output for type T_2 applied to π_I . For any prime p , $\delta_2 = p - 1$, produced as a coordinate reading.

11.2 Justifying γ_2 (SPH §9.2)

Status: Resolved. γ_2 is the identity map on values, with structural interpretation supplied by the type declaration. Multiplicative coherence holds exactly when n is sufficient. The Clifford bilinear form is replaced by the ring/hypercube duality of $\mathbb{Z}/(2^n)\mathbb{Z}$.

11.3 Handling Composite Bases (SPH §9.3)

Status: Resolved. Composite bases introduce no special difficulty because the base is a representational parameter of the type, not a structural parameter of the value. Commutativity follows from datum invariance. Path independence follows from resolver determinism.

12. Testable Predictions

The formalization produces specific testable predictions:

1. **Type constraint monotonicity:** More constrained types produce faster resolution. Confirmed in Phases 2–4: specifying “skewed semiprime” enables $O(1)$ probes; “composite” alone requires $O(N^{1/4})$ rho.
2. **Quantum level sufficiency:** For any finite computation, there exists n such that all operations are exact. The framework operates at arbitrary n with exact arithmetic.
3. **Resolver compositionality:** Type morphisms compose correctly (path independence). Testable by verifying pipelines produce identical results regardless of intermediate types.
4. **Type-relative complexity:** The same datum has different computational complexity under different type declarations. Confirmed: 48-bit composites range from $O(1)$ to $O(N^{1/4})$ depending solely on type constraint.

13. Future Work

1. **Type catalog population:** Define concrete type:TypeDefinition instances for major computational types with content addresses and constraint specifications.
2. **Policy catalog formalization:** Define dispatch rules as a formal function from types to resolvers, with complexity bounds for each binding.
3. **Non-arithmetic types:** Extend to types whose irreducibles are not primes (irreducible polynomials, indecomposable representations, atomic formulas) to validate generality beyond number theory.
4. **Cross-type coherence:** Verify that category $\mathcal{C}$ satisfies universal properties: initial object M_I , limits and colimits, adjunctions between type functors.
5. **Non-arithmetic types:** Extend to types whose irreducibles are not primes (irreducible polynomials, indecomposable representations, atomic formulas) to validate generality beyond number theory.
6. **Observable calibration:** Establish empirical baselines for each of the seven observable families across quantum levels Q0–Q4, quantifying how curvature, cascade length, and monodromy scale with bit width and with type constraint specificity.

Formal specification complete. All definitions grounded in UOR Foundation Ontology v1.0.0. All ten namespaces (u/, schema/, op/, query/, resolver/, type/, observable/, proof/, cert/, derivation/, trace/) mapped to SPH concepts with explicit property bindings. 29 formal definitions, 4 theorems with proofs, 3 open problems resolved constructively. Every ontology class in the foundation graph accounted for: 5 type classes, 4 operation classes, 5 schema classes, 3 query classes, 4 resolver classes, 4 proof classes, 4 certificate classes, 3 derivation classes, 3 trace classes, 13 observable classes, and 2 addressing classes. Verification infrastructure fully specified: every theorem backed by proof:Proof, every transformation by cert:Certificate, every resolution by derivation:Derivation, every path by trace:ComputationTrace, every measurement by observable:Observable. Experimental validation from Phases 1–4 (27 experiments, 260+ composites, 100% accuracy) confirms the type-constraint-to-resolution pipeline.

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